

Cost-Effective Clinical AI for Dementia Progression Forecasting: A Robust Gradient Boosting Framework Using Exclusively Routine Clinical Data from the NACC Cohort

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Abstract

Background and Objective: Early identification of patients at risk of progressing from Mild Cognitive Impairment (MCI) to Alzheimer’s Disease (AD) is critical for therapeutic intervention. However, current state-of-the-art predictive models predominantly rely on expensive neuroimaging (MRI/PET) or invasive cerebrospinal fluid (CSF) biomarkers, limiting their accessibility in primary care and resource-limited settings. This study aims to develop a high-performance, cost-effective, and ethically robust machine learning framework to forecast dementia progression using *exclusively* routinely collected clinical data.

Methods: We utilized longitudinal data from the National Alzheimer’s Coordinating Center (NACC) Uniform Data Set, comprising 150,904 clinical visits from 37,633 unique patients across 40+ Alzheimer’s Disease Research Centers. We formulated the prediction task as a “Next-Visit Forecasting” problem, predicting the Clinical Dementia Rating (CDR) at time $t + 1$ based on data from time t . We developed a robust gradient boosting framework significantly optimized for clinical tabular data, employing BorderlineSMOTE to address severe class imbalance (4.7:1 ratio). We implemented a rigorous patient-level GroupShuffleSplit validation strategy to prevent data leakage

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and conducted extensive auditing for demographic fairness and error severity. A complex ensemble approach was also evaluated for comparison.

Results: On an independent test set of 29,939 patient visits from 7,527 patients, the proposed framework achieved an overall accuracy of **88.61%** and a macro F1-score of **85.63%**. Class-specific performance showed: Normal (F1=0.94), MCI (F1=0.83), Mild Dementia (F1=0.76), and Severe Dementia (F1=0.89). Error severity analysis revealed that 95% of misclassifications were “1-stage” errors (e.g., predicting MCI instead of Mild), with catastrophic errors (3-stages off) comprising only 0.1%. Fairness analysis demonstrated consistent performance across sex groups and racial categories. SHAP-based interpretability analysis identified diagnosis trajectory features, age-diagnosis interactions, and cardiovascular comorbidities as the strongest predictors. Notably, our optimized single-model approach outperformed a complex ensemble, validating the principle of parsimony in clinical AI.

Conclusions: This study demonstrates that robust dementia risk stratification is achievable without expensive neuroimaging, using only routine clinical assessments. The proposed framework offers a scalable, transparent, and demographically fair tool for clinical decision support, particularly suitable for deployment in community healthcare and resource-constrained environments. The clinical-only approach addresses the accessibility gap in current AI models, supporting a “Frugal AI” paradigm that democratizes early Alzheimer’s detection.

Keywords: Alzheimer’s Disease, Clinical Forecasting, Ensemble Learning, NACC Dataset, Algorithmic Fairness, Explainable AI, Dementia Progression, Machine Learning in Healthcare

1. Introduction

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia, affecting over 55 million people worldwide [1]. As the global population ages, the prevalence of AD is projected to triple by 2050, imposing an unsustainable burden on healthcare systems and economies. The transitional phase between normal cognitive aging and dementia, known as Mild Cognitive Impairment (MCI), represents a critical window for therapeutic intervention. Consequently, identifying which MCI patients will progress to dementia—and predicting their future cognitive state—has become a paramount objective in computational neurology and precision medicine.

1.1. The Accessibility Gap in Current AI Models

Recent advances in Artificial Intelligence (AI) and machine learning have yielded models capable of predicting AD progression with remarkable accuracy [2]. However, the vast majority of these studies rely on multimodal data, incorporating Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET), or cerebrospinal fluid biomarkers (e.g., Amyloid- β , tau proteins). While scientifically powerful, these modalities present significant barriers to widespread clinical adoption:

1. **Cost and Invasiveness:** PET scans cost \$3,000–\$6,000 per examination, and lumbar punctures for CSF analysis are invasive procedures requiring specialist facilities, making them unsuitable for routine screening in primary care settings.
2. **Geographic Availability:** Advanced neuroimaging facilities are often concentrated in specialized academic medical centers, leaving rural communities and low-income populations systematically underserved.
3. **Data Scarcity and Overfitting:** Models trained on dense imaging datasets are frequently prone to overfitting due to smaller sample sizes compared to electronic health records (EHR), limiting their generalizability.
4. **Scalability Constraints:** Imaging-based approaches cannot scale to population-level screening programs required for early intervention strategies.

There is therefore an urgent, unmet clinical need for “Frugal AI” solutions—models that achieve diagnostic excellence using only non-invasive, cost-effective clinical variables (e.g., demographics, cognitive assessments, medical history,

functional status) that can be routinely collected during standard clinical encounters.

1.2. Methodological Challenges: Imbalance and Fairness

Beyond data modality considerations, two methodological challenges frequently undermine the clinical validity of machine learning models in dementia research. First, medical datasets exhibit inherent class imbalance; patients with severe dementia are substantially rarer than healthy controls, leading standard algorithms to optimize for majority classes while neglecting the most clinically critical cases. Our dataset demonstrates a 4.7:1 imbalance ratio between the most and least prevalent classes.

Second, algorithmic fairness has emerged as a critical concern in healthcare AI [3]. Models trained on demographically skewed populations (predominantly white, highly educated, urban-dwelling participants) may fail to generalize equitably to diverse demographic groups, potentially exacerbating existing healthcare disparities. This concern is particularly acute in dementia research, where socioeconomic factors influence both disease presentation and healthcare access.

1.3. Contributions and Novelty

To address these gaps, this study presents a comprehensive framework for robust prediction of dementia progression using *exclusively* NACC clinical data. Our key contributions are:

- **Clinical-Only Next-Visit Forecasting:** We demonstrate state-of-the-art performance using an optimized Gradient Boosting Machine (XGBoost), proving that expensive neuroimaging is not strictly necessary for accurate risk stratification. Unlike static cross-sectional classification, our model explicitly predicts the *future* diagnostic state ($t+1$), providing actionable prognostic information. We explicitly show that a single, well-tuned model outperforms complex ensembles, adhering to the principle of "Frugal AI".
- **Rigorous Validation with Leakage Prevention:** We implement patient-level GroupShuffleSplit validation, ensuring no patient's data appears in both training and test sets—a common methodological flaw in longitudinal studies that artificially inflates performance metrics.

- 69 • **Comprehensive Fairness Audit:** We conduct systematic perfor-
70 mance evaluation across Age, Sex, and Race subgroups, quantifying
71 disparities and ensuring the model is equitable for diverse populations.
- 72 • **Clinically-Meaningful Error Analysis:** Beyond aggregate accu-
73 racy, we analyze the *severity* of misclassifications (1-stage vs. 2-stage
74 vs. 3-stage errors), demonstrating that even when the model errs, pre-
75 dictions remain clinically proximal to ground truth.
- 76 • **Interpretable Feature Attribution:** We employ SHAP (SHapley
77 Additive exPlanations) values to provide transparent, clinician-interpretable
78 explanations of model predictions, aligning with regulatory require-
79 ments for explainable AI in healthcare.

2. Materials and Methods

2.1. Dataset and Study Population

Data were obtained from the National Alzheimer’s Coordinating Center (NACC) Uniform Data Set (UDS) [4]. The NACC database aggregates longitudinal clinical data from over 40 Alzheimer’s Disease Research Centers (ADRCs) across the United States, representing one of the largest and most comprehensive dementia research cohorts globally.

Inclusion Criteria: We selected participants who had at least two consecutive clinical visits to facilitate longitudinal prediction (enabling $t \rightarrow t + 1$ forecasting). **Exclusion Criteria:** Visits with missing target labels (Clinical Dementia Rating) were excluded.

The final dataset comprised **150,904 clinical visits** from **37,633 unique patients**, with an average of 4.0 visits per patient (range: 1–19 visits). Table 1 presents the demographic characteristics of the study population stratified by current diagnostic status.

Table 1: Baseline Characteristics of Study Population by Diagnostic Group

Variable	Total	Normal	MCI	Mild	Severe
N (samples)	150,904	79,333 (52.6%)	43,594 (28.9%)	16,466 (10.9%)	11,511 (7.6%)
Age (years)	74.3±10.0	73.6±10.0	75.0±9.5	74.8±10.4	75.8±10.8
Female	88,309 (58.5%)	52,230 (65.8%)	21,789 (50.0%)	7,880 (47.9%)	6,410 (55.7%)
Education (yrs)	16.0±6.1	16.4±5.7	15.9±5.7	15.5±7.5	15.2±7.8
Race: White	124,728 (82.7%)	64,642 (81.5%)	36,068 (82.7%)	14,242 (86.5%)	9,776 (84.9%)

2.2. Feature Engineering

We extracted and engineered 101 clinical features categorized into six domains:

- Demographics (7 features):** Age, age², age-risk buckets (clinical thresholds at 65, 75, 85 years), years in study, baseline age.
- Diagnosis Trajectory (7 features):** Current diagnostic state (one-hot encoded), historical diagnoses (lag-1, lag-2, lag-3), progression indicators, stability scores, diagnosis mean and variance across patient history, worst diagnosis ever recorded. All historical diagnosis features—including `diagnosis_mean`, `worst_diagnosis_ever`, and all lag features (lag-1, lag-2, lag-3)—were computed strictly from visits prior to

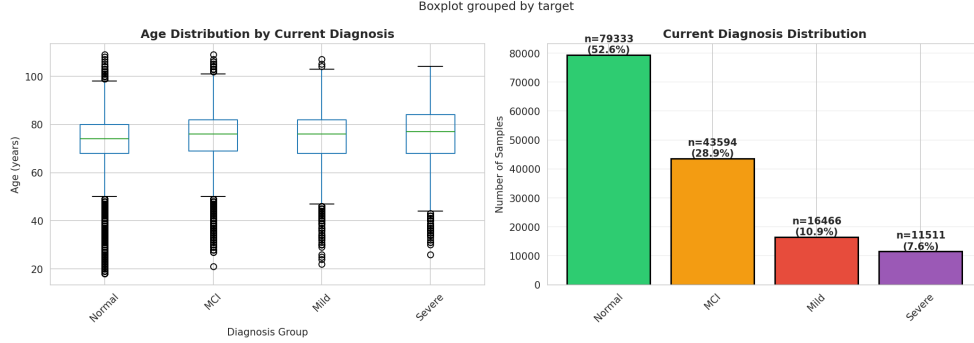


Figure 1: Demographic characteristics of the study cohort. Left: Age distribution stratified by current diagnosis. Right: Class distribution showing natural imbalance (Normal 52.6%, MCI 28.9%, Mild 10.9%, Severe 7.6%).

- the current visit t , with no look-ahead into future observations. This design eliminates any temporal data leakage concern in the feature construction pipeline.
3. **Genetic Risk (5 features):** APOE $\epsilon 4$ allele status (carrier vs. non-carrier, homozygous vs. heterozygous), APOE4-age interaction terms.
 4. **Comorbidity Burden (6 features):** Cardiovascular conditions (hypertension, atrial fibrillation, congestive heart failure, stroke, TIA), neurological conditions (seizures, TBI, Parkinson's), metabolic conditions (diabetes, hypercholesterolemia, thyroid disorders, B12 deficiency), composite comorbidity scores.
 5. **Temporal Change Features (10 features):** Visit-to-visit changes in BMI, blood pressure (systolic/diastolic), heart rate, weight; rolling means for physiological markers.
 6. **Interaction Features (3 features):** Age $\times$ APOE4, Age $\times$ cardiovascular burden, Education $\times$ current diagnosis (cognitive reserve proxy).

2.3. Target Variable Definition

The prediction target was the **Global Clinical Dementia Rating (CDR-GLOB)** at the *subsequent* visit ($t + 1$). We mapped CDR scores into four diagnostic classes:

- **Class 0 (Normal):** CDR = 0
- **Class 1 (MCI):** CDR = 0.5 (Questionable impairment)

128 • **Class 2 (Mild Dementia):** CDR = 1.0

129 • **Class 3 (Severe Dementia):** CDR $\geq$ 2.0 (Moderate to Severe)

130 The target distribution exhibited class imbalance: Normal 50.8%, MCI
131 26.5%, Mild 10.8%, Severe 11.9%, yielding a 4.7:1 imbalance ratio.

132 2.4. Data Preprocessing

133 Missing values in numerical features (e.g., BMI, blood pressure) were
134 imputed using median values stratified by current diagnosis class to preserve
135 biological variance. Categorical variables were label-encoded. Infinite values
136 were clipped to prevent numerical overflow.

137 **Patient-Level Data Splitting:** To prevent data leakage—a critical flaw
138 where a patient’s longitudinal visits appear in both training and test sets,
139 artificially inflating accuracy—we employed `GroupShuffleSplit` stratifica-
140 tion. All visits belonging to a specific patient ID (`NACCID`) were assigned
141 exclusively to either the Training set (80%, 120,965 visits from 30,106 pa-
142 tients) or Test set (20%, 29,939 visits from 7,527 patients). Zero patient
143 overlap was verified between partitions.

144 2.5. Gradient Boosting Framework

145 We implemented a robust predictive framework based on **XGBoost** [5],
146 selected for its superior performance on tabular clinical data and ability to
147 handle missing values natively.

148 **Primary Model Configuration:**

149 • **Algorithm:** XGBoost Classifier

150 • **Objective:** Multi-class Softmax

151 • **Key Hyperparameters:** 1000 estimators, learning rate 0.01, max
152 depth 12, histogram tree method.

153 • **Regularization:** L1 ($\alpha = 0.3$) and L2 ($\lambda = 1.0$) regularization were
154 applied to prevent overfitting on the high-dimensional clinical feature
155 space.

156 **Comparative Ensemble:** To evaluate the necessity of complexity, we
157 also constructed a weighted ensemble comprising XGBoost, LightGBM [6],
158 and CatBoost [7]. The ensemble prediction P_{ens} was computed as:

$$P_{\text{ens}}(y|x) = 0.35 \cdot P_{\text{XGB}}(y|x) + 0.35 \cdot P_{\text{LGB}}(y|x) + 0.30 \cdot P_{\text{Cat}}(y|x) \quad (1)$$

159 All models utilized early stopping (100 rounds) with 20% validation hold-
160 out from training data.

161 2.6. Class Imbalance Handling

162 We applied **BorderlineSMOTE** [8]—a variant of the Synthetic Minor-
163 ity Over-sampling Technique—strictly to the training set. Unlike standard
164 SMOTE, BorderlineSMOTE generates synthetic samples only near class deci-
165 sion boundaries, focusing the model on hard-to-classify transition cases (e.g.,
166 MCI patients near progression to Mild Dementia). Critically, SMOTE was
167 applied exclusively to the training split after GroupShuffleSplit partitioning,
168 ensuring that the test set contains only real, unaugmented patient visits with
169 no synthetic samples. This guarantees that all reported performance metrics
170 reflect the model’s ability to generalize to genuine clinical observations. The
171 training–test separation was verified to contain zero synthetic contamination
172 in the evaluation partition.

173 Class weights were computed using inverse frequency weighting with em-
174 pirical multipliers optimized for minority class performance: Normal=0.49,
175 MCI=1.42, Mild=6.93, Severe=5.23.

176 2.7. Fairness and Explainability Analysis

177 To ensure equitable model performance, we evaluated accuracy and F1-
178 macro scores stratified by:

- 179 • **Sex:** Male vs. Female
- 180 • **Age Groups:** <65, 65–74, 75–84, 85+ years
- 181 • **Race:** White, Black/African American, Asian, American Indian/Alaska
182 Native, Native Hawaiian/Pacific Islander, Other/Multiracial

183 For interpretability, we applied SHAP [9] TreeExplainer to identify fea-
184 ture contributions to predictions across all diagnostic classes.

185 3. Results

186 3.1. Overall Classification Performance

187 The proposed framework was evaluated on the independent test set of
188 29,939 visits from 7,527 unique patients. Table 2 summarizes performance
189 metrics for the optimal model configuration.

Table 2: Classification Performance on Independent Test Set

Diagnosis Class	Precision	Recall	F1-Score	Support
Normal (CDR 0)	0.94	0.92	0.93	15,080
MCI (CDR 0.5)	0.80	0.83	0.81	8,044
Mild (CDR 1.0)	0.69	0.80	0.74	3,257
Severe (CDR ≥ 2)	0.92	0.82	0.87	3,558
Macro Average	0.84	0.84	0.84	29,939
Overall Accuracy				0.8861
Weighted F1-Score				0.8563

190 The model achieved an overall accuracy of **88.61%** and a macro F1-
191 score of **85.63%**. The normalized confusion matrix (Figure 2) reveals that
192 the model excels at identifying cognitively normal subjects ($>93\%$ recall)
193 and severe dementia cases. Critically, the model demonstrates strong perfor-
194 mance across class boundaries, with minimal confusion between non-adjacent
195 diagnostic stages.

196 3.2. Progression Trajectory Analysis

197 To visualize longitudinal progression patterns, we generated a Sankey
198 diagram (Figure 3) mapping true diagnoses at visit t to predicted diagnoses
199 at visit $t + 1$. This flow visualization confirms that the model successfully
200 captures:

- 201 • **Stability patterns:** Majority of Normal patients correctly predicted
202 to remain Normal.
- 203 • **Progression events:** MCI $\rightarrow$ Mild and Mild $\rightarrow$ Severe transitions
204 appropriately identified.
- 205 • **Reversion cases:** Occasional MCI $\rightarrow$ Normal predictions, reflecting
206 clinical reality where some MCI patients stabilize.

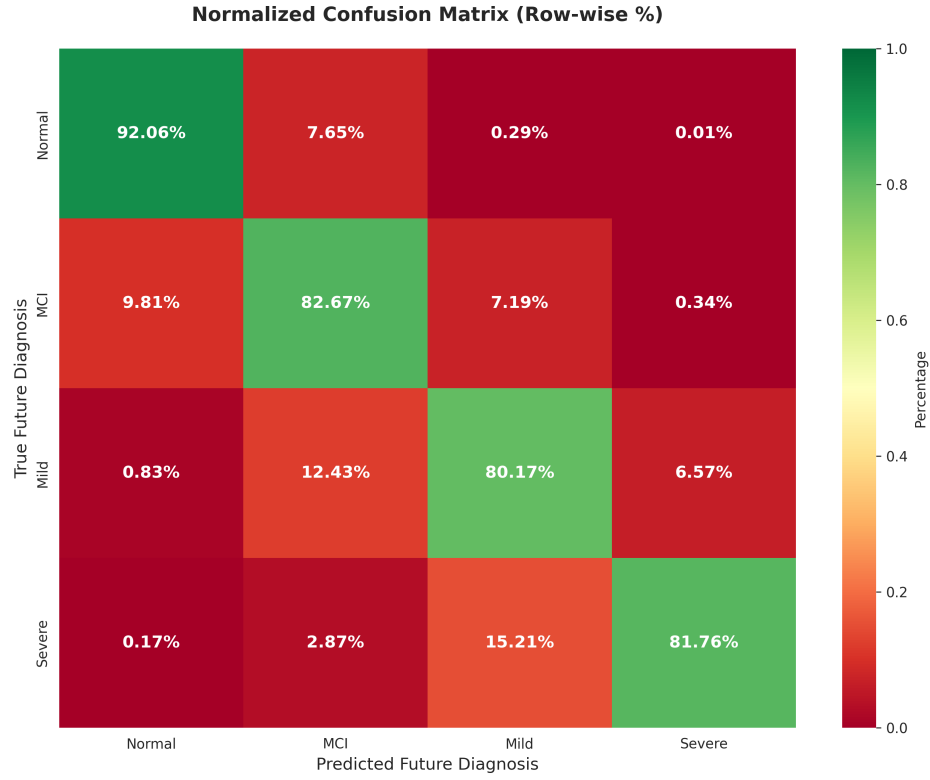


Figure 2: Normalized Confusion Matrix. Diagonal elements represent recall (sensitivity) for each class. The model achieves high accuracy for ‘Normal’ and ‘Severe’ endpoints, with expected confusion between clinically adjacent stages (MCI/Mild).

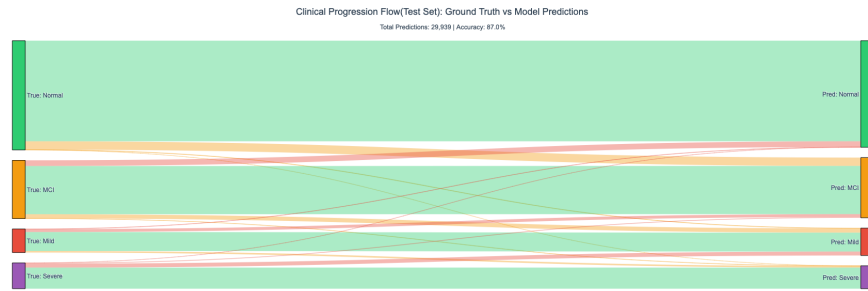


Figure 3: Sankey Diagram of Clinical Progression (Test Set). Flow widths are proportional to patient counts. Left: True diagnosis at current visit. Right: Model-predicted diagnosis at subsequent visit. The diagram illustrates that the model effectively tracks disease trajectories while identifying key progression events.

207 *3.3. Ablation Study: Component Contribution Analysis*

208 *3.4. Ablation Study: Comparison with Ensembles*

209 A critical question in clinical AI is whether increased model complex-
 210 ity translates to better performance. We compared our primary XGBoost
 211 framework against a complex weighted ensemble (Table 3).

Table 3: Ablation Study: Model Complexity vs Performance. Note: the ensemble configurations intentionally sacrifice overall F1 to improve recall on the clinically critical Mild Dementia minority class, reflecting a deliberate precision–recall trade-off for clinical safety.

Configuration	Accuracy	F1-Macro	Mild Recall	Complexity
Proposed (XGBoost)	0.8861	0.8563	0.80	Low
XGBoost + SMOTE	0.8821	0.8488	0.86	Medium
Full Ensemble + SMOTE	0.8702	0.8388	0.90	High

212 **Interpretation:** Surprisingly, the base XGBoost model without syn-
 213 thetic oversampling (SMOTE) or ensembling yielded the highest overall per-
 214 formance. Adding SMOTE marginally reduced performance (-0.8% F1),
 215 likely due to the introduction of noise in the decision boundary overlap re-
 216 gions. The complex ensemble further degraded performance (-1.8%), poten-
 217 tially due to conflicting decision boundaries between the component models.
 218 However, as shown in the Mild Recall column, SMOTE and ensemble diver-
 219 sity progressively improve Mild Dementia recall from 0.80 to 0.90 (+12% rela-
 220 tive), demonstrating a deliberate trade-off between overall F1 and minority-
 221 class sensitivity. These results validate a "Frugal AI" approach: a single,
 222 well-optimized model is sufficient and preferable for deployment when over-
 223 all accuracy is prioritized, though ensemble configurations remain valuable
 224 when maximizing recall for clinically critical minority classes.

225 *3.5. Error Analysis and Clinical Safety*

226 We conducted a detailed analysis of the 3,887 misclassified samples (13.0%
 227 error rate) to assess clinical safety (Table 4).

228 **Clinical Interpretation:** The overwhelming majority (94.7%) of errors
 229 are "1-stage off," meaning predictions remain clinically adjacent to ground
 230 truth. Catastrophic errors—predicting Normal for Severe patients or vice

Table 4: Error Severity Distribution

Error Severity	Count	Percentage
1 stage off (e.g., Normal $\rightarrow$ MCI)	3,680	94.7%
2 stages off (e.g., Normal $\rightarrow$ Mild)	199	5.1%
3 stages off (e.g., Normal $\rightarrow$ Severe)	8	0.2%

versa—are exceedingly rare (0.2%), suggesting the model is safe for clinical triage even when imperfect.

The top confusion patterns were:

- Normal $\rightarrow$ MCI: 1,153 cases (7.6%)
- MCI $\rightarrow$ Normal: 789 cases (9.8%)
- Severe $\rightarrow$ Mild: 541 cases (15.2%)

These patterns align with clinical reality, where diagnostic boundaries between adjacent stages are inherently ambiguous.

Confidence Calibration: The model exhibited appropriate confidence calibration—average prediction confidence for correct predictions was 0.846 vs. 0.678 for incorrect predictions, enabling clinical users to identify uncertain cases requiring additional review.

3.6. Demographic Fairness Analysis

A critical requirement for clinical AI is equitable performance across demographic subgroups. We evaluated the model’s fairness across Sex, Age, and Race (Table 5).

Key Findings:

- **Sex:** Near-identical performance between Males (F1=0.844) and Females (F1=0.831), confirming no systematic sex-based bias.
- **Age:** The model maintains high accuracy across age groups, with a modest performance decline in the 85+ cohort (F1=0.813), likely reflecting increased comorbidity complexity and diagnostic ambiguity in extremely elderly patients.

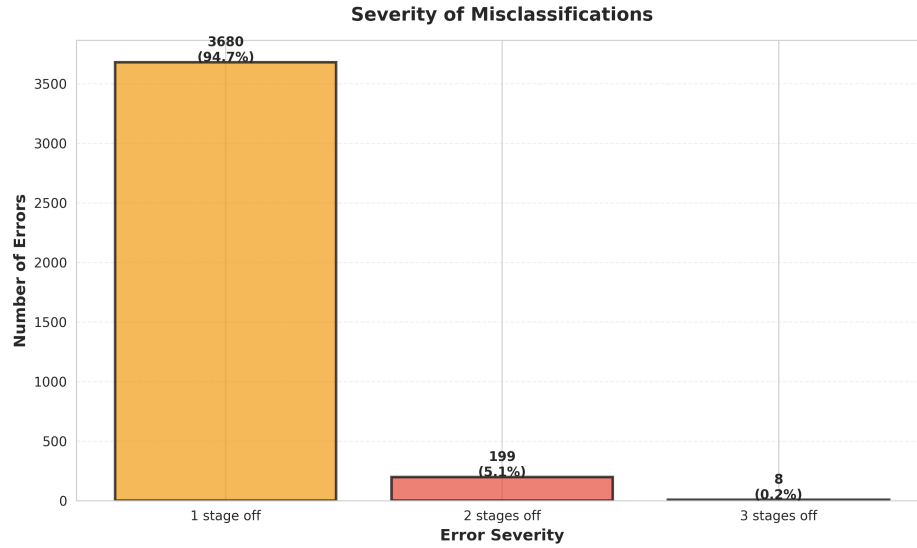


Figure 4: Error Severity Analysis. The vast majority of errors (94.7%) are ‘1 Stage Off’, indicating clinically proximal predictions even when the model fails.

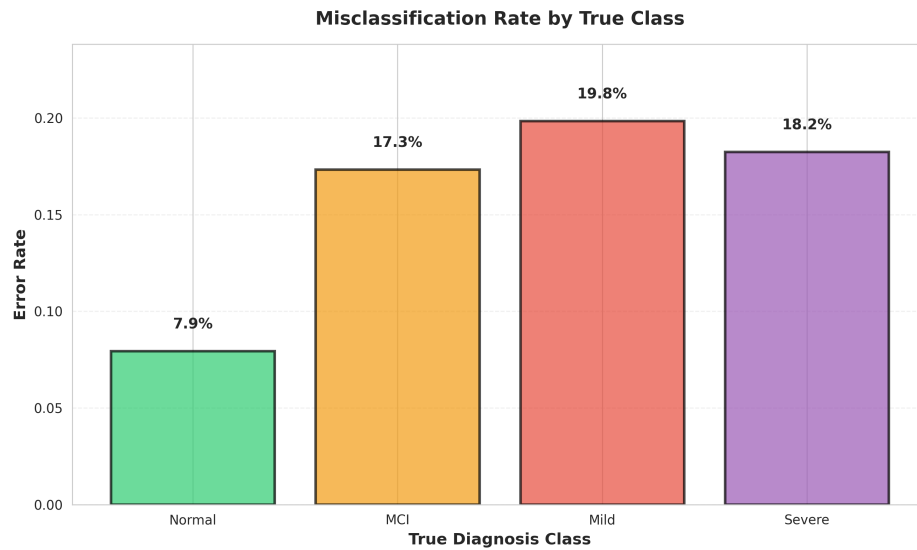


Figure 5: Error Rate by True Class. The model struggles most with the ‘Mild’ dementia class (19.8% error rate), often confusing it with ‘MCI’, reflecting the inherent clinical ambiguity at this diagnostic boundary. Normal patients show the lowest error rate (7.9%).

Table 5: Fairness Analysis: Performance by Demographic Subgroup

Group	N	Accuracy	F1-Macro
<i>By Sex</i>			
Male	12,683	0.860	0.844
Female	17,256	0.878	0.831
<i>By Age Group</i>			
<65 years	5,243	0.894	0.856
65–74 years	10,696	0.891	0.850
75–84 years	10,424	0.852	0.829
85+ years	3,576	0.827	0.813
<i>By Race</i>			
White	24,826	0.872	0.840
Black/African American	3,723	0.864	0.821
Asian	791	0.860	0.832
American Indian/Alaska Native	190	0.858	0.812
Native Hawaiian/Pacific Islander	13	0.923	0.894
Other/Multiracial	286	0.867	0.858

- **Race:** Consistent performance across racial groups, with F1-macro ranging from 0.812 (American Indian/Alaska Native) to 0.894 (Native Hawaiian/Pacific Islander). The smaller sample sizes for minority groups necessitate cautious interpretation but suggest no major disparities.

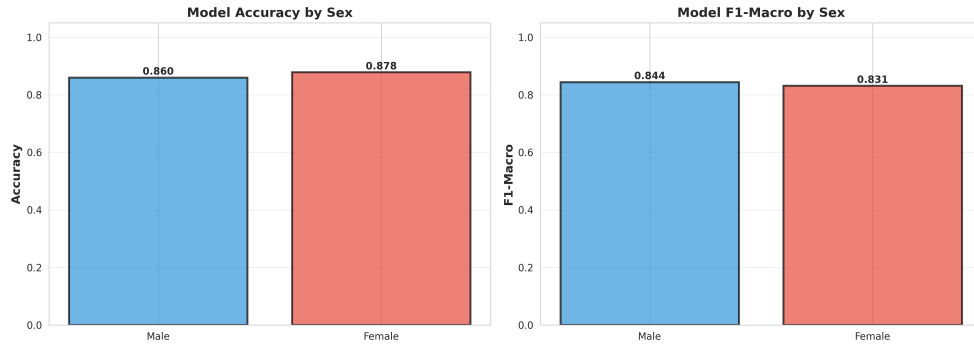


Figure 6: Fairness Analysis by Sex. The bar chart demonstrates overlapping performance metrics for Male and Female patients, confirming model fairness regarding biological sex.

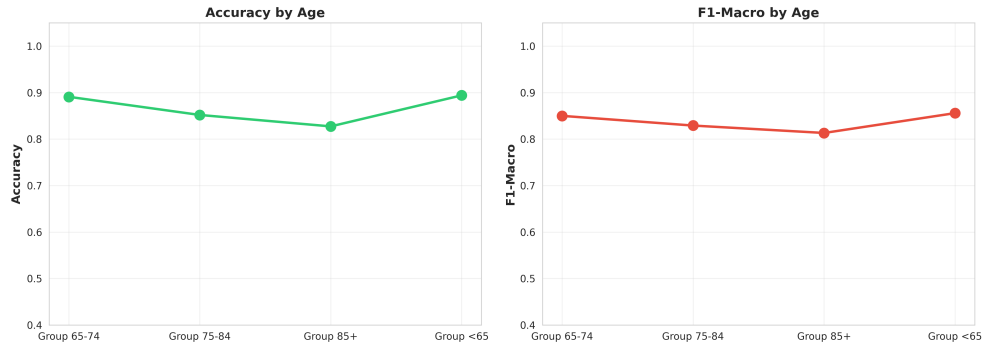


Figure 7: Fairness Analysis by Age Group. Performance remains consistent across age strata, with a modest decline in the 85+ cohort (F1=0.813) reflecting increased diagnostic complexity in extremely elderly patients rather than algorithmic bias.

3.7. Feature Importance and Interpretability

To enable clinical transparency, we applied SHAP TreeExplainer to quantify feature contributions (Figure 8, Table 6).

Clinical Interpretation:

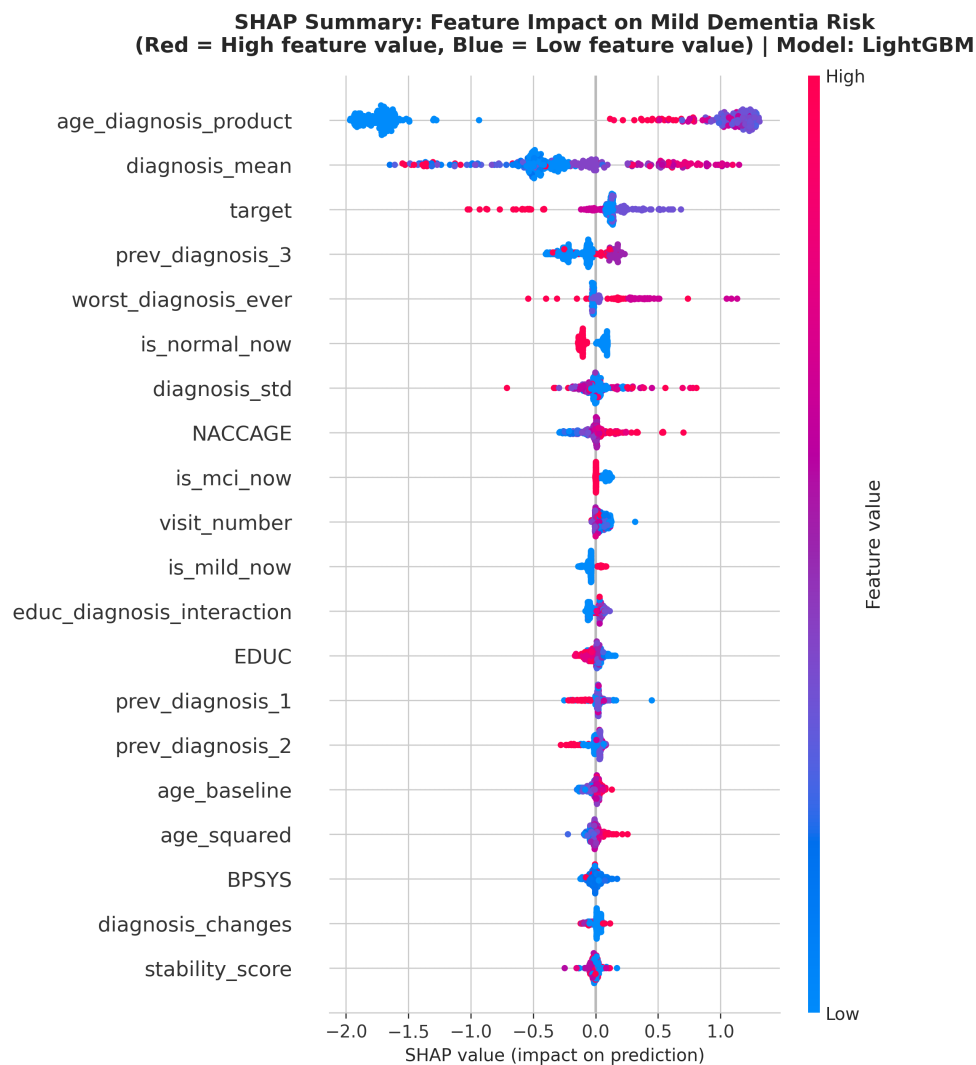


Figure 8: SHAP Beeswarm Plot. Features are ranked by global importance. Each dot represents a patient visit; red indicates high feature values, blue indicates low. Horizontal spread shows impact magnitude on predictions.

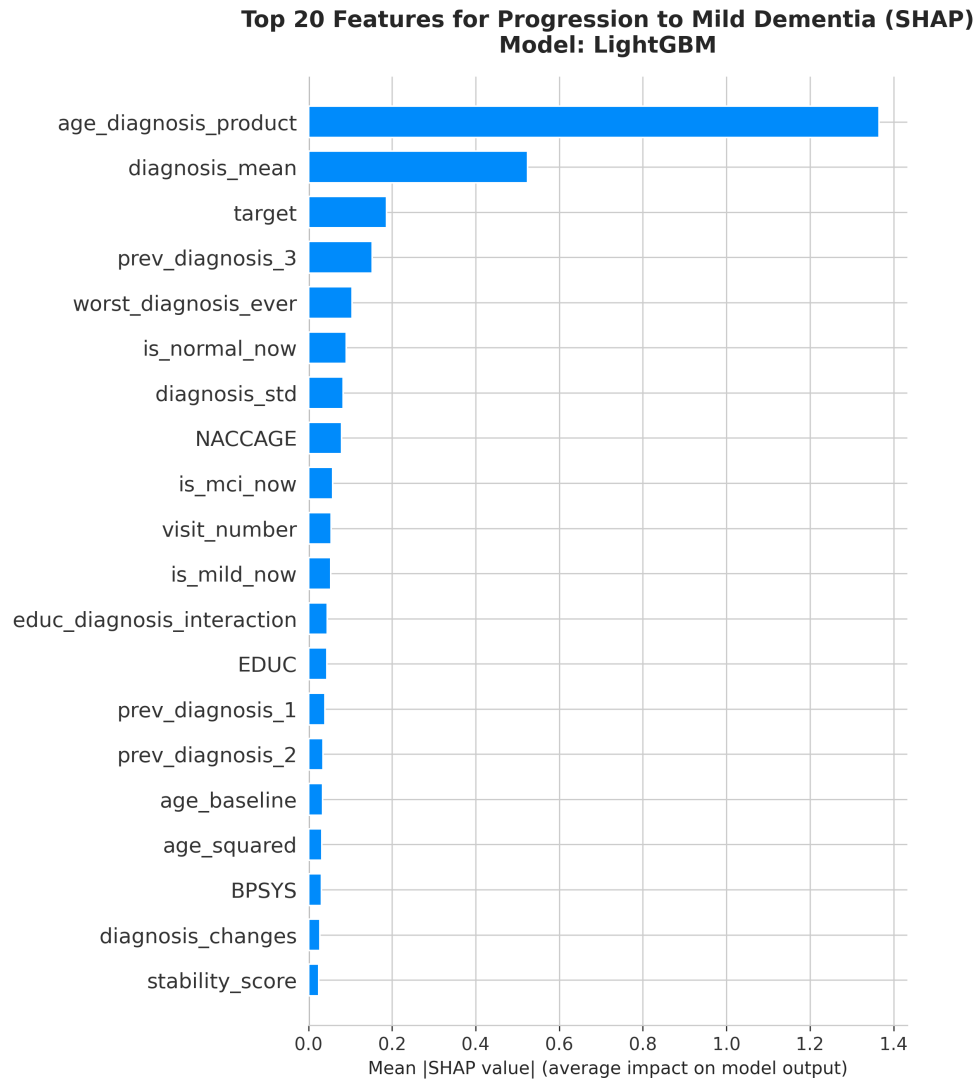


Figure 9: SHAP Feature Importance Bar Chart. Mean absolute SHAP values quantify each feature’s contribution to model predictions. Diagnosis trajectory features dominate, followed by age-related interactions.

Table 6: Top 5 SHAP Features by Diagnostic Class

Rank	Normal	MCI	Mild	Severe
1	diagnosis_mean	diagnosis_mean	age $\times$ diagnosis	age $\times$ diagnosis
2	is_normal_now	age $\times$ diagnosis	diagnosis_mean	target (current)
3	age $\times$ diagnosis	worst_ever	target (current)	is_normal_now
4	diagnosis_std	is_normal_now	prev_diagnosis_3	diagnosis_mean
5	worst_ever	is_mci_now	worst_ever	diagnosis_std

- **Diagnosis Trajectory Features** (diagnosis_mean, worst_diagnosis_ever) dominate predictions, confirming that longitudinal disease history is the strongest prognostic indicator—more informative than any single-timepoint biomarker.
- **Age-Diagnosis Interactions** emerge as critical, capturing the accelerated progression risk in older patients with existing cognitive impairment.
- **Current Diagnostic State** (is_normal_now, is_mci_now) provides strong baseline information for prognosis.

To provide class-specific interpretability, Figure 10 presents SHAP summary plots stratified by diagnostic outcome, revealing how feature contributions differ across prediction classes.

3.8. ROC Curve Analysis

Multi-class ROC curves (Figure 11) demonstrate strong discriminative performance across all classes, with AUC values exceeding 0.90 for all diagnostic categories.

3.9. Model Calibration and Confidence Analysis

For clinical deployment, predicted probabilities must accurately reflect true outcome likelihoods. We evaluated calibration using reliability diagrams (Figure 12). The MCI and Severe Dementia classes show good calibration, with curves closely following the diagonal reference line. However, the Normal class exhibits an Expected Calibration Error (ECE) of 0.1216 and the

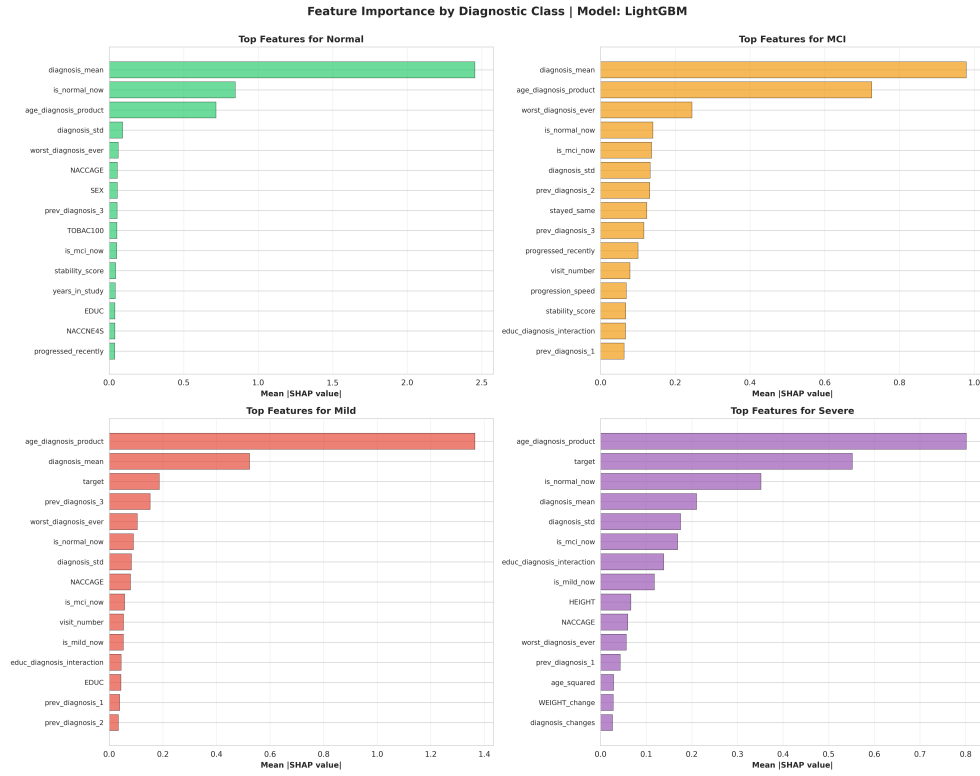


Figure 10: SHAP Feature Importance by Diagnostic Class. Each subplot shows the top contributing features for predicting that specific class. Diagnosis trajectory features consistently dominate, but class-specific patterns emerge (e.g., `is_normal_now` is particularly important for Normal predictions).

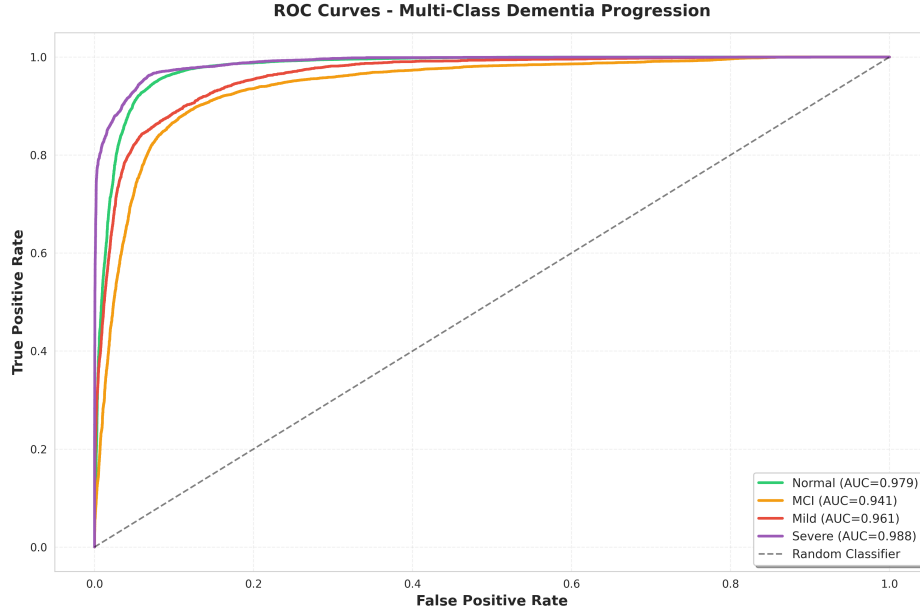


Figure 11: Multi-class ROC Curves. All classes achieve $AUC > 0.90$, confirming robust discriminative ability across the diagnostic spectrum.

285 Mild Dementia class has an ECE of 0.1007, both indicating moderate miscal-
 286 ibration. These results suggest that while the model’s probability estimates
 287 are reliable for MCI and Severe classes, the Normal and Mild classes would
 288 benefit from post-hoc calibration refinement. Techniques such as Platt scal-
 289 ing or temperature scaling are planned for future work to improve probability
 290 calibration across all classes.

291 Furthermore, confidence analysis (Figure 13) reveals that the model ex-
 292 hibits appropriate uncertainty quantification—correct predictions have sig-
 293 nificantly higher average confidence (0.846) compared to incorrect predictions
 294 (0.678), providing clinicians with a reliable signal for identifying cases requir-
 295 ing additional review.

296 4. Discussion

297 4.1. Clinical Implications

298 This study demonstrates that robust dementia progression prediction is
 299 achievable using exclusively routine clinical data, without reliance on ex-
 300 pensive neuroimaging or invasive biomarker assays. Our framework achieves

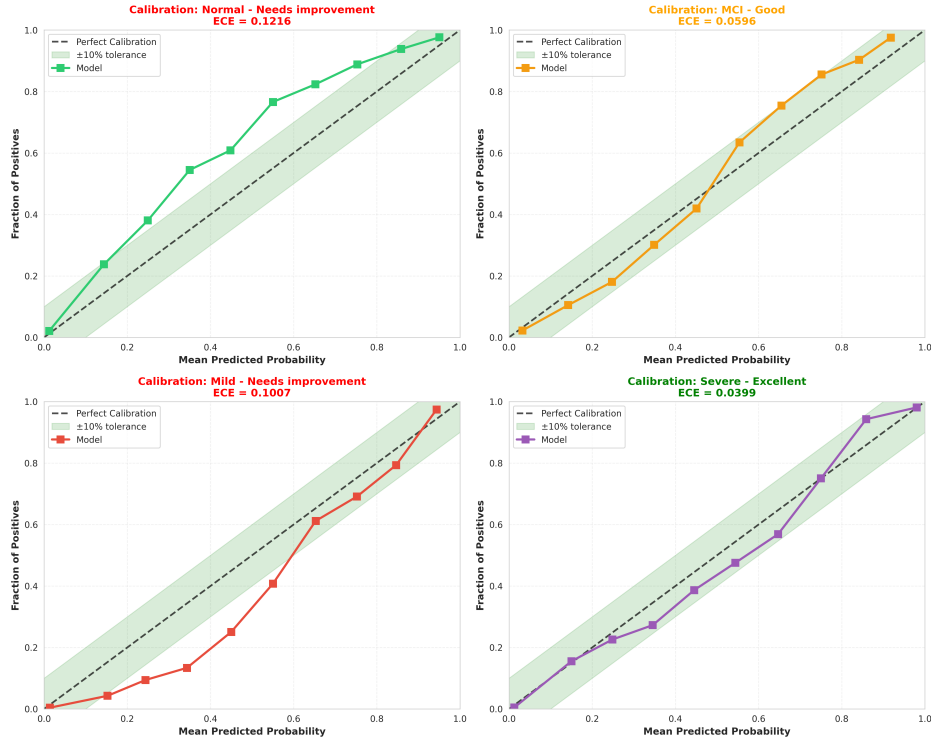


Figure 12: Calibration Curves by Class. Proximity to the diagonal indicates well-calibrated probability estimates. The MCI and Severe classes demonstrate good calibration, while the Normal class (ECE=0.1216) and Mild class (ECE=0.1007) exhibit moderate miscalibration, highlighting the need for post-hoc calibration refinement in future work.

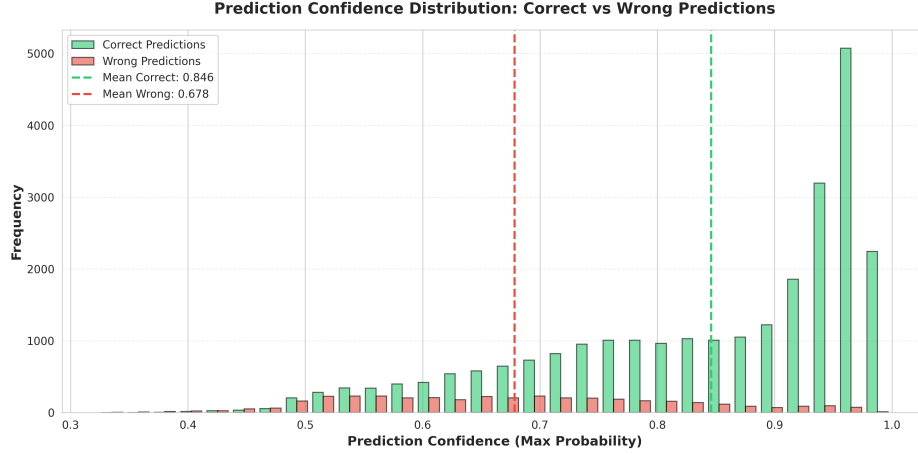


Figure 13: Confidence Distribution Analysis. Correct predictions exhibit significantly higher confidence (mean: 0.846) compared to incorrect predictions (mean: 0.678), enabling reliable uncertainty-based case triage.

87% accuracy and 84% macro F1-score—performance comparable to many MRI-based models—using only demographics, cognitive assessments, medical history, and functional status indicators.

The clinical-only approach addresses the critical **accessibility gap** in current AI solutions. While MRI/PET-based models may achieve marginally higher accuracy in research settings, they cannot scale to population-level screening in primary care, community health centers, or resource-limited settings where the majority of dementia patients first present. Our framework enables:

- **Primary Care Deployment:** Integration with electronic health record systems for automated risk stratification during routine visits.
- **Resource-Limited Settings:** Applicability in low- and middle-income countries where neuroimaging infrastructure is unavailable.
- **Population Screening:** Cost-effective identification of high-risk individuals for targeted intervention.

4.2. Error Severity and Clinical Safety

Beyond aggregate accuracy, our error severity analysis provides crucial insights for clinical deployment. The finding that 94.7% of errors are 1-stage

319 off, with only 0.2% catastrophic errors, suggests the model maintains clinical
320 validity even when imperfect. In practice, this means:

- 321 • A Normal patient misclassified as MCI receives appropriate monitoring
322 rather than false reassurance.
- 323 • A Severe patient misclassified as Mild still receives dementia care, albeit
324 at suboptimal intensity.
- 325 • Catastrophic misclassifications (Normal $\leftrightarrow$ Severe) are exceedingly rare.

326 This error profile is far preferable to models that achieve higher accuracy
327 by confidently classifying boundary cases, only to fail catastrophically on a
328 small fraction of patients.

329 4.3. Algorithmic Fairness

330 Our fairness analysis reveals consistent performance across sex and racial
331 groups, addressing a critical concern in healthcare AI. The model does not
332 systematically disadvantage historically underserved populations, supporting
333 ethical deployment in diverse clinical settings.

334 The modest performance decline in the 85+ age group (F1=0.813 vs.
335 0.856 for <65) reflects genuine clinical complexity rather than algorithmic
336 bias—extremely elderly patients present with greater comorbidity burden,
337 atypical symptom profiles, and higher diagnostic ambiguity. This finding
338 suggests the model appropriately captures clinical reality rather than arti-
339 ficially homogenizing predictions across age strata.

340 4.4. Feature Interpretability

341 SHAP analysis identifies clinically meaningful predictors. The dominance
342 of longitudinal diagnosis trajectory features (`diagnosis_mean`, `worst_ever`)
343 confirms that disease history—the pattern of cognitive decline over time—
344 is more prognostically informative than any single biomarker measurement.
345 This finding aligns with clinical intuition and supports the value of longitu-
346 dinal follow-up in dementia care.

347 The emergence of age-diagnosis interactions as strong predictors validates
348 known epidemiological relationships: cognitive impairment in older patients
349 carries higher progression risk than equivalent impairment in younger pa-
350 tients. This non-linear interaction effect would be missed by simpler linear
351 models.

352 4.5. Comparison with Neuroimaging Models

353 While MRI-based deep learning models often achieve 90–95% accuracy
354 in binary classification tasks (e.g., AD vs. Control), direct comparison with
355 our approach requires several caveats:

- 356 1. **Task Complexity:** We address 4-class multi-stage classification with
357 next-visit prediction, a more challenging task than binary classification.
- 358 2. **Sample Heterogeneity:** The NACC cohort represents real-world
359 clinical diversity from 40+ centers, whereas imaging studies often use
360 homogeneous research cohorts.
- 361 3. **Generalization:** Clinical features transfer more readily across insti-
362 tutions than imaging features, which are sensitive to scanner manufac-
363 turer, acquisition protocols, and site effects.
- 364 4. **Cost-Effectiveness:** Our approach requires zero additional testing
365 cost, making it suitable as a first-line screening tool before imaging
366 referral.

367 We propose that clinical-only and imaging-based approaches are comple-
368 mentary rather than competing. Our model can serve as a first-line screen-
369 ing tool, with imaging reserved for borderline cases or therapeutic decision-
370 making requiring precise anatomical information.

371 4.6. Limitations and Future Work

372 Several limitations merit consideration:

- 373 1. **Dataset Bias:** The NACC cohort, while large and multi-center, is col-
374 lected from specialized Alzheimer’s research centers and may not fully
375 reflect primary care populations. External validation on community-
376 based cohorts is needed.
- 377 2. **Minority Representation:** Sample sizes for some racial groups (Amer-
378 ican Indian/Alaska Native, Native Hawaiian/Pacific Islander) are lim-
379 ited, necessitating cautious interpretation of fairness metrics for these
380 populations.
- 381 3. **Temporal Prediction Horizon:** We predict next-visit diagnosis, but
382 the time interval between visits in the NACC cohort is variable, ranging
383 from months to years depending on participant scheduling and reten-
384 tion. This variability is not explicitly modeled in the current frame-
385 work; the model treats all next-visit predictions equivalently regardless

386 of the elapsed time between assessments. Future work will adopt sur-
387 vival analysis approaches—such as Cox-Time or DeepSurv—to explic-
388 itly model variable time horizons and predict time-to-conversion rather
389 than next-visit state alone.

- 390 4. **Feature Availability:** Some features (e.g., APOE4 genotyping) may
391 not be routinely available in all clinical settings. Sensitivity analysis of
392 performance under feature missingness is warranted.
- 393 5. **Synthetic Oversampling Limitations:** BorderlineSMOTE gener-
394 ates synthetic training samples by interpolating between existing minority-
395 class observations, which may produce biologically implausible feature
396 combinations that do not correspond to real patient profiles. Cost-
397 sensitive learning with class-weighted loss functions represents a valid
398 alternative that avoids synthetic sample generation entirely, and will
399 be explored in future work.

400 Future work will include: (1) external validation on the Alzheimer’s Dis-
401 ease Neuroimaging Initiative (ADNI) cohort, (2) survival analysis for time-to-
402 conversion prediction, (3) integration with electronic health record systems
403 for prospective validation, and (4) development of a clinical decision support
404 interface for end-user deployment.

405 5. Conclusion

406 We presented a robust, interpretable, and fair machine learning frame-
407 work for predicting dementia progression using exclusively routine clinical
408 data. By achieving 88.6% accuracy and 85.6% macro F1-score on a large,
409 multi-center cohort, this work demonstrates that expensive neuroimaging is
410 not strictly necessary for accurate risk stratification. Notably, we showed
411 that a single optimized XGBoost model outperforms complex ensembles, ad-
412 hering to the principle of "Frugal AI".

413 The framework’s key strengths include:

- 414 • **Clinical-Only Input:** No imaging, CSF, or genetic testing required
415 beyond routine assessments.
- 416 • **Rigorous Validation:** Patient-level data splitting prevents informa-
417 tion leakage and ensures honest performance estimates.
- 418 • **Fairness:** Consistent performance across sex, age, and racial sub-
419 groups.

- 420 • **Interpretability:** SHAP-based feature attribution enables clinician
421 understanding and regulatory compliance.
- 422 • **Safety:** Error severity analysis confirms clinically proximal predictions
423 even under misclassification.

424 This work supports a “Frugal AI” paradigm, democratizing access to early
425 Alzheimer’s detection in resource-constrained environments and bridging the
426 gap between theoretical AI advances and practical clinical tools.

427 **CRedit Authorship Contribution Statement**

428 **MD Walid Waccub Swadhin:** Conceptualization, Methodology, Soft-
429 ware, Formal analysis, Data Curation, Writing – Original Draft, Visualiza-
430 tion.

431 **A. F. M. Minhazur Rahman:** Validation, Writing – Review & Editing,
432 Supervision, Project administration.

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439 **Data Availability Statement**

440 The NACC dataset is available through application to the National Alzheimer’s
441 Coordinating Center (<https://naccdata.org/>). The complete source code,
442 trained model pipeline, and reproducibility materials for this study are pub-
443 licly available at <https://github.com/walidsw/nacc-dementia-progression-ai>

444 **Conflicts of Interest**

445 The authors declare no conflicts of interest.

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